

Course Code: CAP 770	Course Instructor: Dr. Uminder Kaur
Course Title: Advance Data Structures	Roll Number: RD2110B61
Name: Atul Kumar	Registration Number: 12102801

Question: Program to create a graph using linked list.

Answer: -

```
#include<iostream>

using namespace std;

struct node
{
    int data;
    struct node *link;
};

struct vertexlist
{
    struct node *vlisthead;
};

struct Graph
{
    int v;
    struct vertexlist *vl; };
```

```
struct Graph* CreateGraph(int n)  
{  
    int i;  
    struct Graph *vlist = new Graph;  
    vlist->v = n;  
    vlist->vl = new vertexlist[n+1];  
        for(i = 0; i < n+1; i++)  
        {  
            vlist->vl[i].vlisthead = NULL;  
        }  
    return vlist;  
}  
  
struct node* NewNode(int value)  
{  
    struct node *newnode = new node;  
    newnode->data = value;  
    newnode->link = NULL;  
    return newnode;  
}
```

```
void InsertNode(Graph *G, int v1, int v2)  
{  
    node *newnode1 = NewNode(v1);  
    node *newnode2 = NewNode(v2);  
    node *temp = new node;  
    if(G->vl[v2].vlisthead == NULL)  
    {  
        G->vl[v2].vlisthead = newnode1;  
    }  
    else  
    {  
        newnode1->link = G->vl[v2].vlisthead;  
        G->vl[v2].vlisthead = newnode1;  
    }  
    if(G->vl[v1].vlisthead == NULL)  
    {  
        G->vl[v1].vlisthead = newnode2;  
    }  
    else
```

```

    {
        newnode2->link = G->vl[v1].vlisthead;
        G->vl[v1].vlisthead = newnode2;
    }
}

int main()
{
    int i, v, e;

    cout<<"Enter the number of vertexes of the
graph: ";

    cin>>v;

    struct Graph *G = CreateGraph(v);

    cout<<"\nEnter the number of edges of the
graph: ";

    cin>>e;

    int edge[e][2];

    for(i = 0; i < e; i++)
    {
        cout<<"\nEnter the vertex pair for edge "<<i+1;

```

```

        cout<<"\nV(1): ";
        cin>>edge[i][0];
        cout<<"V(2): ";
        cin>>edge[i][1];
        InsertNode(G, edge[i][0], edge[i][1]);
    }

    cout<<"\n\nThe incidence list representation for
the given graph: ";
    for(i = 0; i < v; i++)
    {
        cout<<"\n\tV("<i+1<<" ) -> { ";
        while(G->vl[i+1].vlisthead != NULL)
        {
            cout<<G->vl[i+1].vlisthead->data<<"
";
            G->vl[i+1].vlisthead = G-
>vl[i+1].vlisthead->link;
        }
        cout<<"}";
    }

```

}

}

```
Enter the number of vertexes of the graph: 2
```

```
Enter the number of edges of the graph: 2
```

```
Enter the vertex pair for edge 1
```

```
V(1): 2
```

```
V(2): 3
```

```
Enter the vertex pair for edge 2
```

```
V(1): 4
```

```
V(2): 5
```

```
The incidence list representation for the given graph:
```

```
V(1) -> { }
```

```
V(2) -> { 3 }
```

```
...Program finished with exit code 0
```

```
Press ENTER to exit console.█
```



